

- Window Shoppers

Business Questions Answer about the Window Shoppers are as follows:

- 1) What percentage of our total walk-in visitors left the store without making a purchase, and how significant is the window shopper problem for the business?
- 2) Which age group accounts for the highest number of window shoppers, and does a particular generation show a stronger tendency to browse without buying?
- 3) Are New customers or Referral customers more likely to leave without making a purchase, and what does this reveal about the quality of leads coming through each acquisition channel?
- 4) Is there a pattern between a customer's vision type and their likelihood of being a window shopper, and does the type of eye condition influence the purchase decision?
- 5) Which staff members are handling the highest number of customers who leave without purchasing, and does this indicate a gap in sales conversion skills at the individual staff level?

Why this analysis matters

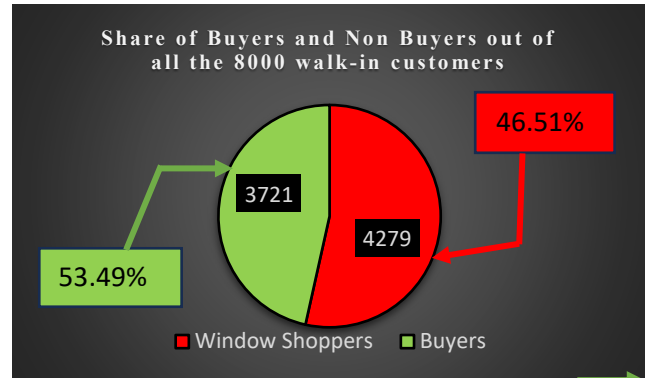
Window shoppers represent customers who visited the store but did not make a purchase. Analysing this segment helps identify lost sales opportunities, evaluate the effectiveness of the sales process, and uncover customer groups with lower conversion rates. Understanding why customers leave without buying enables the business to improve customer experience, strengthen sales strategies, and increase overall conversion rates.

1) Count of Window Shoppers out of Total Visited Customers

1) What's percentage of our total walk-in visitors left the store without making a purchase, and how significant is the window shopper problem for the business?

```
SELECT
COUNT(DISTINCT c.customer_id) AS Total_Visited_Customers,
COUNT(DISTINCT CASE WHEN i.customer_id IS NULL THEN c.customer_id END) AS Window_Shoppers,
COUNT(DISTINCT CASE WHEN i.customer_id IS NOT NULL THEN c.customer_id END) AS Buyers,
ROUND(COUNT(DISTINCT CASE WHEN i.customer_id IS NULL THEN c.customer_id END) * 100 /
COUNT(DISTINCT c.customer_id), 2) AS Percentage_of_Window_Shoppers,
ROUND(COUNT(DISTINCT CASE WHEN i.customer_id IS NOT NULL THEN c.customer_id END) * 100 /
COUNT(DISTINCT c.customer_id), 2) AS Percentage_Buyers
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id;
```

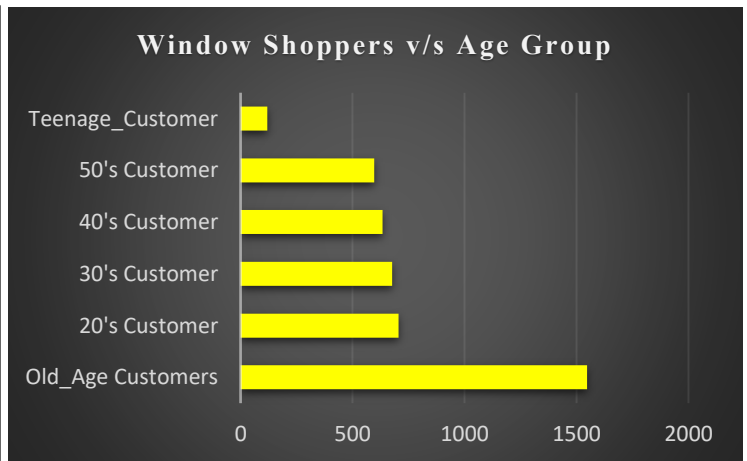
Total_Visited_Customers	Window_Shoppers	Buyers	Percentage_of_Window_Shoppers	Percentage_Buyers
8000	4279	3721	53.49	46.51



2) Count of Window Shoppers across all Age Groups

2) Which age group accounts for the highest number of window shoppers, and does a particular generation show a stronger tendency to browse without buying?

```
SELECT
CASE
WHEN c.age < 20 THEN "Teenage_Customer"
WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"
WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"
WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"
WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"
WHEN c.age > 60 THEN "Old_Age Customers"
END AS Age_group,
COUNT(DISTINCT c.customer_id)
AS Total_Visited_Customers,
COUNT(DISTINCT CASE WHEN i.customer_id IS NULL THEN c.customer_id END)
AS COUNT_Window_Shoppers,
ROUND(COUNT(DISTINCT CASE WHEN i.customer_id IS NULL THEN c.customer_id END)*100/
COUNT(DISTINCT c.customer_id), 2) AS Percentage_of_Window_Shoppers
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY Age_group WITH ROLLUP
ORDER BY COUNT_Window_Shoppers DESC;
```



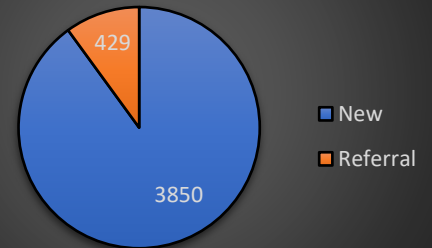
Age_group	Total_Visited_Customers	COUNT_Window_Shoppers	Percentage_of_Window_Shoppers
NULL	8000	4279	53.49
Old_Age Customers	2920	1547	52.98
20's Customer	1328	705	53.09
30's Customer	1214	677	55.77
40's Customer	1157	634	54.80
50's Customer	1152	597	51.82
Teenage_Customer	229	119	51.97

3) Count of Window Shoppers across different Customer Type

- 3) Are New customers or Referral customers more likely to leave without making a purchase, and what does this reveal about the quality of leads coming through each acquisition channel?

```
SELECT c.customer_type,
COUNT(DISTINCT c.customer_id )
AS Total_Visited_Customers,
COUNT(DISTINCT i.customer_id)
AS Buyers,
COUNT(DISTINCT CASE WHEN i.customer_id IS NULL THEN c.customer_id END)
AS Window_Shoppers,
ROUND(COUNT(DISTINCT CASE WHEN i.customer_id IS NULL THEN c.customer_id END)*100/
COUNT(DISTINCT c.customer_id ), 2)
AS Percentage_of_Window_Shoppers
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
GROUP BY c.customer_type;
```

Count of Window Shoppers across customer type



Result Grid

Filter Rows:

Export:

Wrap Cell Content:

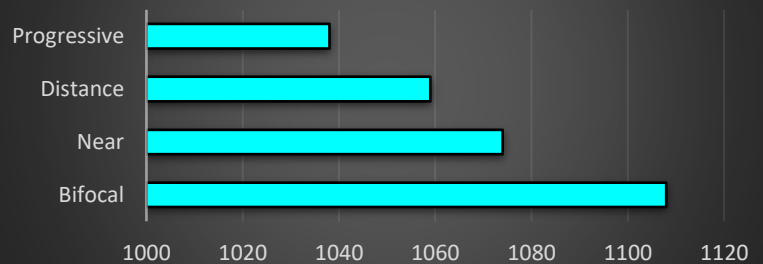
	customer_type	Total_Visited_Customers	Buyers	Window_Shoppers	Percentage_of_Window_Shoppers
▶	New	7190	3340	3850	53.55
	Referral	810	381	429	52.96

4) Count of Window Shoppers across their Vision Type

- 4) Is there a pattern between a customer's vision type and their likelihood of being a window shopper, and does the type of eye condition influence the purchase decision?

```
SELECT vision_type, COUNT(c.customer_id)
AS COUNT_Window_Shoppers
FROM customers_details as c
LEFT JOIN prescriptions as p
ON c.customer_id=p.customer_id
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
WHERE i.customer_id IS NULL
GROUP BY vision_type
ORDER BY COUNT_Window_Shoppers DESC;
```

Count of Window Shoppers across Vision Type



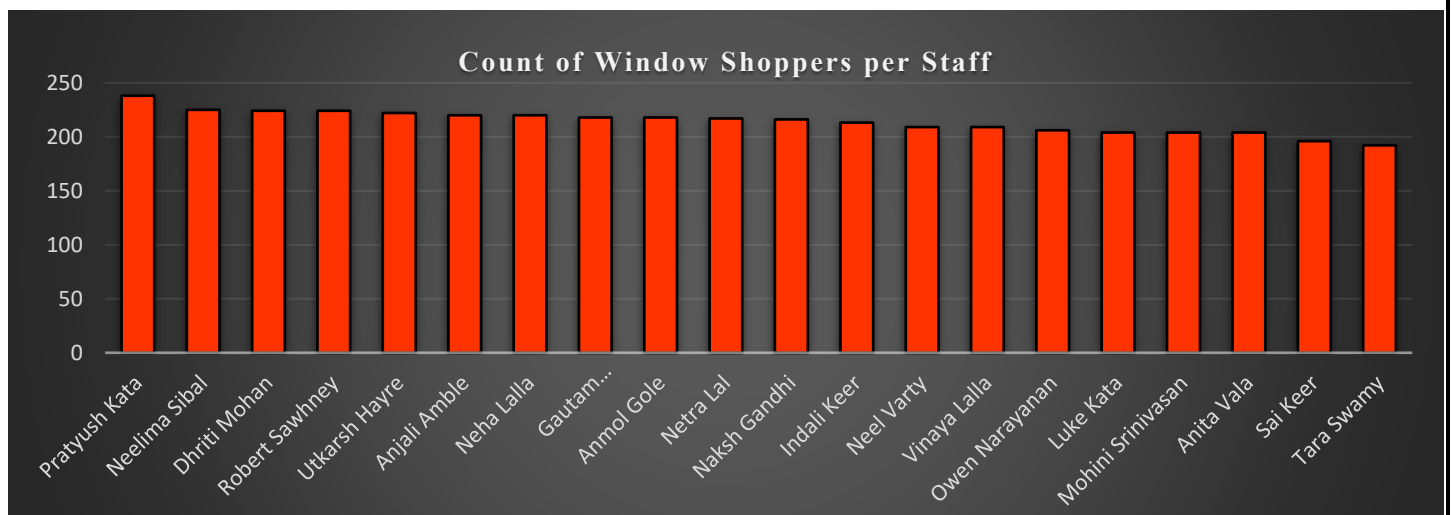
Result Grid	Filter Rows:
vision_type	COUNT_Window_Shoppers
Bifocal	1108
Near	1074
Distance	1059
Progressive	1038

5) Count of Window Shoppers per Staff

5) Which staff members are handling the highest number of customers who leave without purchasing, and does this indicate a gap in sales conversion skills at the individual staff level?

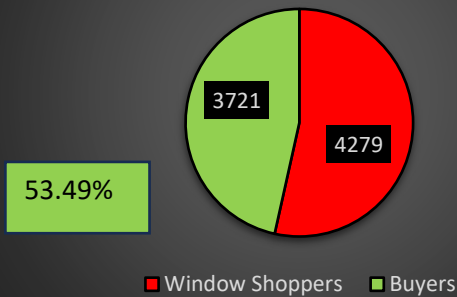
```
SELECT s.staff_id,
       s.staff_full_name,
       COUNT(c.customer_id)
       AS COUNT_Window_Shoppers
FROM customers_details as c
LEFT JOIN staff as s
ON c.assigned_staff_id=s.staff_id
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
WHERE i.customer_id IS NULL
GROUP BY s.staff_id, s.staff_full_name
ORDER BY COUNT_Window_Shoppers DESC;
```

Result Grid	Filter Rows:	Export
staff_id	staff_full_name	COUNT_Window_Shoppers
16	Pratyush Kata	238
10	Neelima Sibal	225
5	Dhriti Mohan	224
7	Robert Sawhney	224
20	Utkarsh Hayre	222
2	Anjali Amble	220
9	Neha Lalla	220
14	Gautam Choudhary	218
17	Anmol Gole	218
15	Netra Lal	217
12	Naksh Gandhi	216
13	Indali Keer	213
1	Neel Varty	209
4	Vinaya Lalla	209
18	Owen Narayanan	206
3	Luke Kata	204
8	Mohini Srinivasan	204
19	Anita Vala	204
6	Sai Keer	196

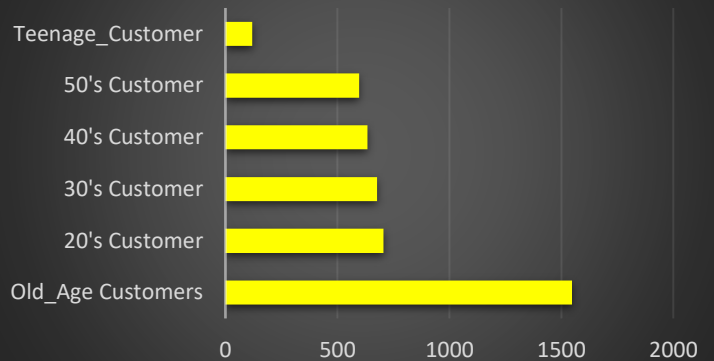


Window Shoppers Dashboard

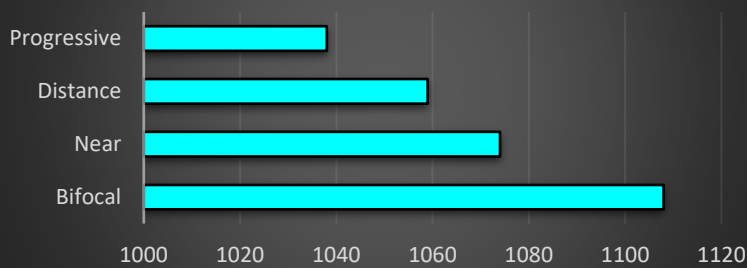
Share of Buyers and Non Buyers out of all the 8000 walk-in customers



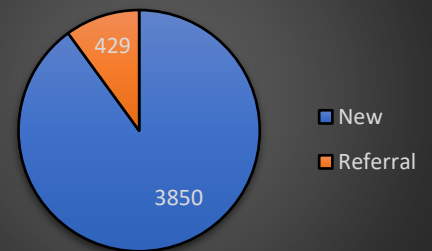
Window Shoppers v/s Age Group



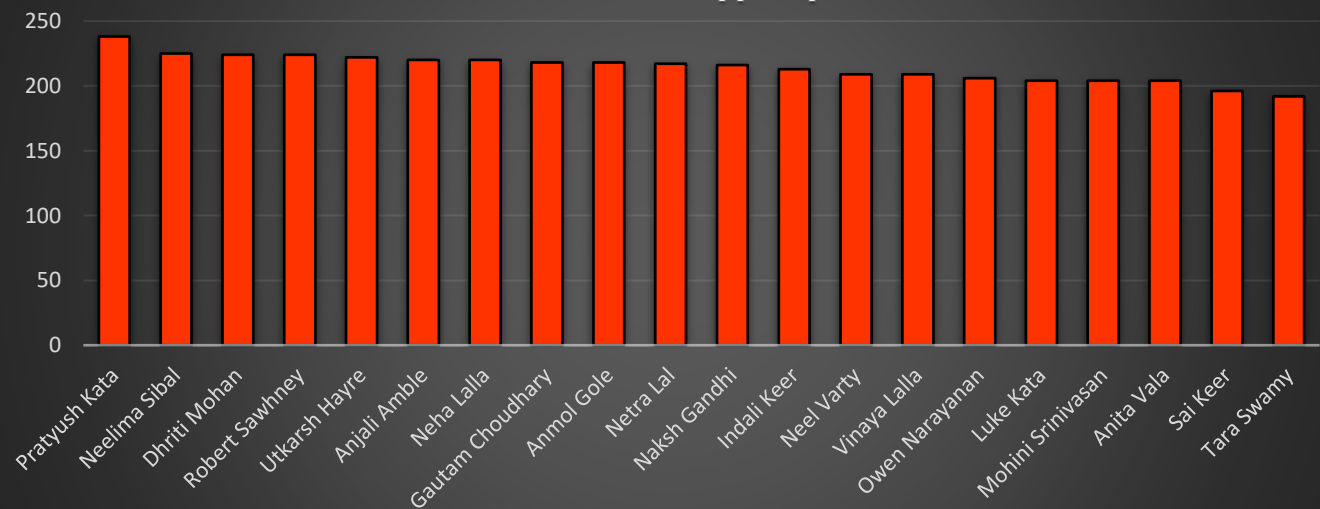
Count of Window Shoppers across Vision Type



Count of Window Shoppers across customer type



Count of Window Shoppers per Staff



- 4.2.3) Actual Buyer Customers

Business Questions Answer about the Actual Buyer Customers are as follows:

Please Note: This section uses two distinct measures depending on the nature of the business question are as follows:

- Unique Buyers (3,721) is used for all customer-level questions such as conversion rate, age group distribution, customer type, and vision type ensuring each customer is counted only once regardless of how many times they purchased.
- Total Transactions (5,000) is used for all product-level questions such as product performance, brand preference, and product type where every individual purchase transaction is a meaningful and separate business event.

This distinction ensures accuracy and prevents data inflation at every level of analysis.

- 1) Out of all the customers who walked into our store, how many unique individuals actually made a purchase and what does our overall conversion rate reveal about the effectiveness of our sales process?
- 2) Which age group contributes the highest number of actual buyers, and how does the buyer distribution across age groups compare to the overall walk-in distribution?
- 3) Which customer type New or Referral has a higher proportion of actual buyers, and does being a referral customer make someone more likely to make a purchase?
- 4) Which vision type Near, Distance, Bifocal, or Progressive is most commonly associated with customers who actually make a purchase, and does the severity or type of eye condition drive purchase intent?
- 5) Which products are most frequently purchased by our actual buyers, and which product category drives the highest sales volume in the store?
- 6) Which brands are most preferred by our converted customers, and which brand drives the highest revenue among actual buyers?
- 7) Which product types are most commonly bought by our converted customers, and does product type preference vary across different buyer segments?

Why this analysis matters

Buyer customers are the primary source of the store's revenue. Analysing this segment helps identify which customer groups generate the highest sales, contribute the most revenue, and demonstrate stronger purchasing behaviour. These insights enable the business to better understand its core customer base, optimize marketing efforts, improve sales performance, and allocate resources toward the most valuable customer segments.

1) Count of Unique Real Customers

1) Out of 8,000 total walk-in customers, how many unique individuals actually made a purchase and what does our overall conversion rate reveal about the effectiveness of our sales process?

```
SELECT
COUNT(DISTINCT c.customer_id) AS Total_Visited_Customers,
COUNT(DISTINCT CASE WHEN i.customer_id IS NOT NULL THEN i.customer_id END)
AS COUNT_Unique_Real_Customers,
ROUND( COUNT(DISTINCT CASE WHEN i.customer_id IS NOT NULL THEN i.customer_id END) * 100 /
COUNT(DISTINCT c.customer_id), 2) AS Conversion_Rate
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id;
```

	Total_Visited_Customers	COUNT_Unique_Real_Customers	Conversion_Rate
▶	8000	3721	46.51

Share of Buyer Customers
out of all the customers



■ Total Visted Customers ■ Unique Customers

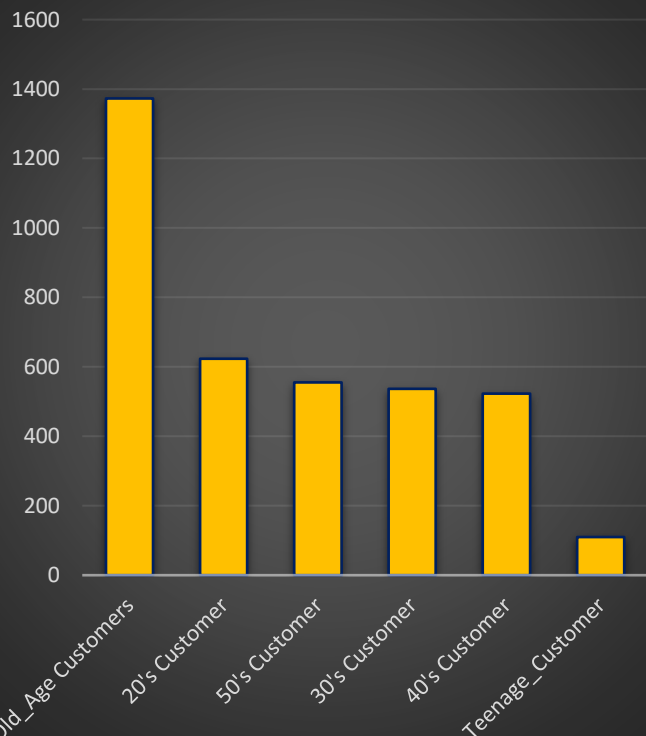
2) Count of Buyer Customers across different Age Group

2) Which age group contributes the highest number of actual buyers, and how does the buyer distribution across age groups compare to the overall walk-in distribution?

```
SELECT
CASE
WHEN c.age < 20 THEN "Teenage_Customer"
WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"
WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"
WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"
WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"
WHEN c.age > 60 THEN "Old_Age Customers"
END AS Age_group,
COUNT(DISTINCT i.customer_id) AS COUNT_Repeated_Customers
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
WHERE i.customer_id IS NOT NULL
GROUP BY Age_group
ORDER BY COUNT_repeated_customers DESC;
```

	Age_group	COUNT_Repeated_Customers
▶	Old_Age Customers	1373
	20's Customer	623
	50's Customer	555
	30's Customer	537
	40's Customer	523
	Teenage_Customer	110

Buyer Customers



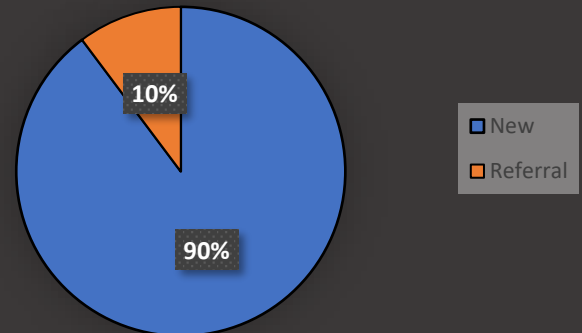
3) Buyer Customers across different Customer Type

3) Which customer type New or Referral has a higher proportion of actual buyers, and does being a referral customer make someone more likely to make a purchase?

```
SELECT
    c.customer_type,
    COUNT(DISTINCT i.customer_id)
    AS Count_Customers
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
WHERE i.customer_id IS NOT NULL
GROUP BY c.customer_type;
```

Result Grid	Filter Rows:
customer_type	Count_Customers
New	3340
Referral	381

Buyer Customers across different customers type



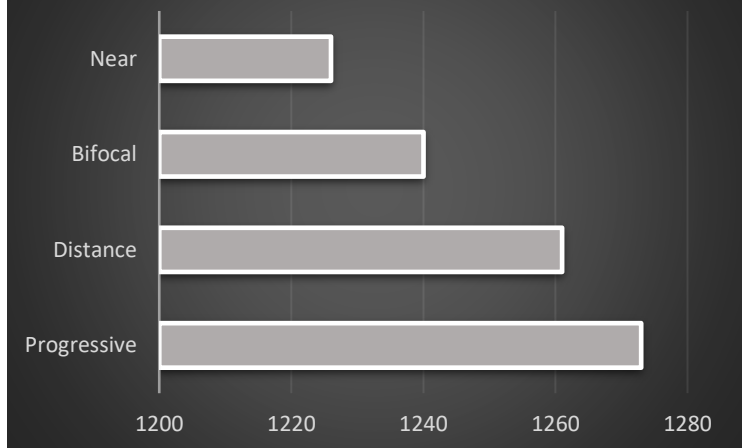
4) Buyer Customers and their Vision Types

4) Which vision type Near, Distance, Bifocal, or Progressive is most commonly associated with customers who actually make a purchase, and does the severity or type of eye condition drive purchase intent?

```
SELECT p.vision_type, COUNT(i.customer_id)
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
LEFT JOIN prescriptions as p
ON c.customer_id=p.customer_id
WHERE i.customer_id IS NOT NULL
GROUP BY p.vision_type
ORDER BY COUNT(i.customer_id) DESC;
```

Result Grid	Filter Rows:
vision_type	COUNT(i.customer_id)
Progressive	1273
Distance	1261
Bifocal	1240
Near	1226

Buyers Customers v/s Vision Type

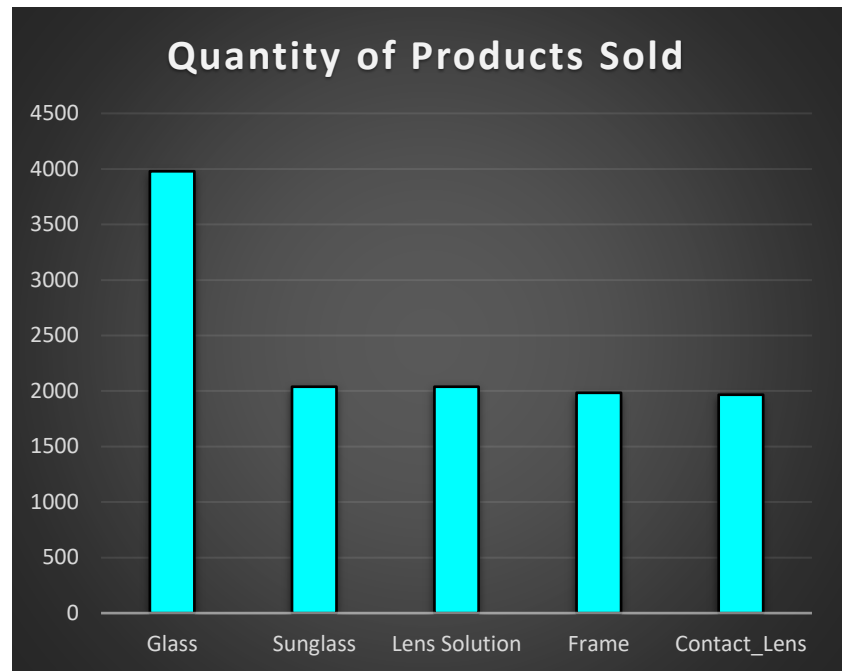


5) Highest Buying Products by Buyer Customers

5) Which products are most frequently purchased by our actual buyers, and which product category drives the highest sales volume in the store?

```
SELECT
  p.product_id,
  p.product_name,
  SUM(ii.quantity) AS Count_Products
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
LEFT JOIN invoice_items as ii
ON i.invoice_id=ii.invoice_id
LEFT JOIN products as p
ON ii.product_id=p.product_id
WHERE i.customer_id IS NOT NULL
GROUP BY p.product_id, p.product_name
ORDER BY Count_Products DESC;
```

Result Grid			Filter Rows:
	product_id	product_name	Count_Products
▶	2	Glass	3980
	3	Sunglass	2040
	6	Lens Solution	2038
	1	Frame	1984
	4	Contact_Lens	1968



6) QTY Sold per Brand across Buyer Customers

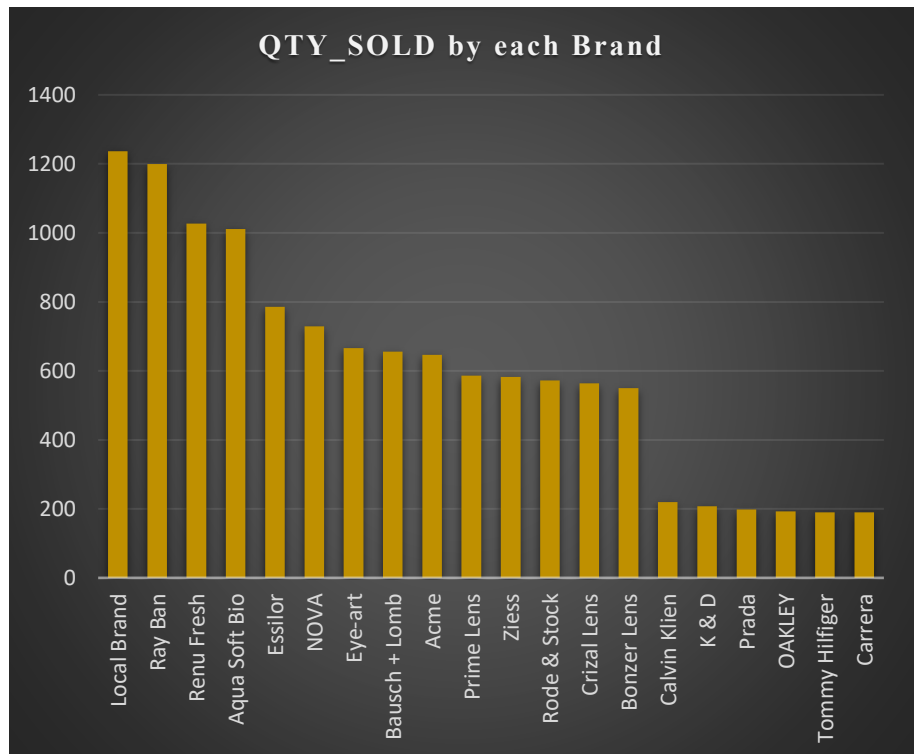
6) Which brands are most preferred by our converted customers, and which brand drives the highest revenue among actual buyers?

```
WITH customers AS (
SELECT DISTINCT c.customer_id
FROM customers_details as c
),
customers_invoices AS (
SELECT cc.customer_id, i.invoice_id
FROM customers as cc
INNER JOIN invoices as i
ON cc.customer_id=i.customer_id
),
customers_invoices_items AS (
SELECT
ci.customer_id, ci.invoice_id,
ii.item_id, ii.product_id,
ii.brand_id, ii.product_type_id,
ii.quantity
FROM customers_invoices as ci
INNER JOIN invoice_items as ii
ON ci.invoice_id=ii.invoice_id
),
```

```
customers_invoices_products_name AS (
SELECT
cii.customer_id, cii.invoice_id,
cii.item_id, cii.product_id,
cii.brand_id, cii.product_type_id,
cii.quantity, pd.product_name
FROM customers_invoices_items as cii
INNER JOIN products as pd
ON cii.product_id=pd.product_id
),
customers_invoices_products_name_brand_name AS (
SELECT
cipn.customer_id, cipn.invoice_id,
cipn.item_id, cipn.product_id,
cipn.brand_id, cipn.product_type_id,
cipn.quantity, cipn.product_name,
pb.brand_name
FROM customers_invoices_products_name as cipn
INNER JOIN product_brand as pb
ON cipn.brand_id=pb.brand_id
)
```

```
SELECT
cipnbn.product_name,
cipnbn.brand_name,
SUM(cipnbn.quantity) AS QTY_SOLD
FROM customers_invoices_products_name_brand_name
as cipnbn
GROUP BY cipnbn.product_name, cipnbn.brand_name
ORDER BY QTY_SOLD DESC;
```

brand_name	QTY_SOLD
Local Brand	1237
Ray Ban	1199
Renu Fresh	1027
Aqua Soft Bio	1011
Essilor	786
NOVA	729
Eye-art	666
Bausch + Lomb	656
Acme	646
Prime Lens	586
Ziess	582
Rode & Stock	572
Crizal Lens	564
Bonzer Lens	550
Calvin Klien	220
K & D	208
Prada	198
OAKLEY	193
Tommy Hilfiger	190
Carrera	190

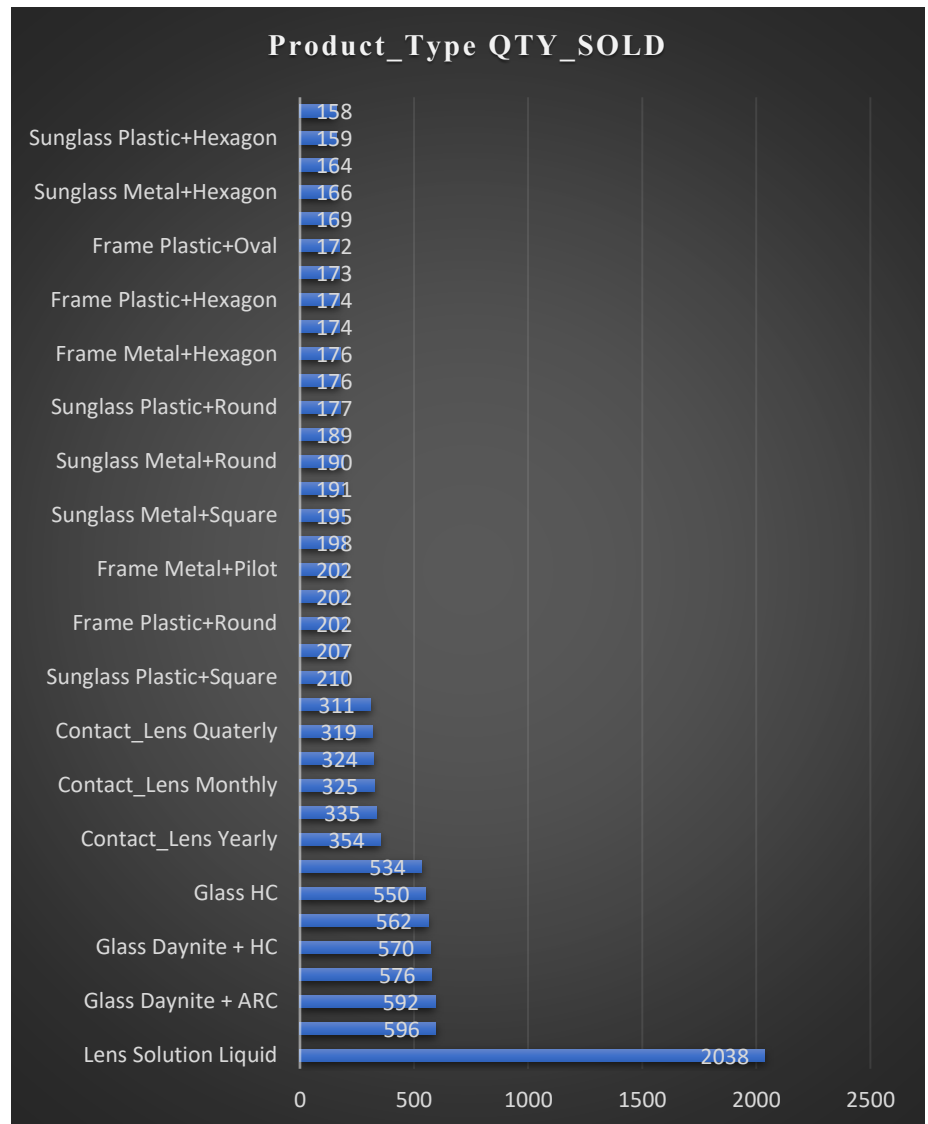


7) QTY Sold for each product type

7) Which product types are most commonly bought by our converted customers, and does product type preference vary across different buyer segments?

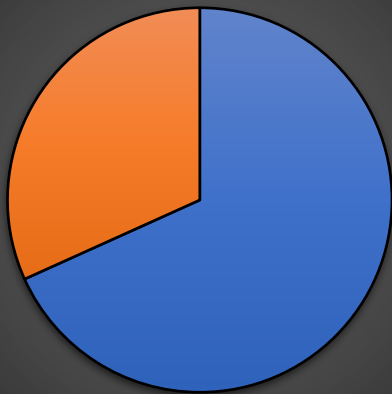
```
SELECT
    pd.product_name, cpbtqr.product_type_id,
    product_type_name, SUM(cpbtqr.quantity)
    AS QTY_SOLD
FROM customers_product_brand_type_qty_revenue
as cpbtqr
INNER JOIN product_type AS pt
ON cpbtqr.product_type_id=pt.product_type_id
INNER JOIN products as pd
ON pd.product_id=cpbtqr.product_id
GROUP BY pd.product_name,
    cpbtqr.product_type_id,
    product_type_name
ORDER BY QTY_SOLD DESC;
```

Result Grid	Filter Rows:	Export:	Wrap
product_name	product_type_id	product_type_name	QTY_SOLD
Lens Solution	36	Liquid	2038
Glass	13	ARC	596
Glass	17	Daynite + ARC	592
Glass	18	Daynote + Blu-Block	576
Glass	16	Daynite + HC	570
Glass	15	Daynite	562
Glass	12	HC	550
Glass	14	Blu-Block	534
Contact_Lens	35	Yearly	354
Contact_Lens	30	One day	335
Contact_Lens	32	Monthly	325
Contact_Lens	31	Weekly	324
Contact_Lens	33	Quarterly	319
Contact_Lens	34	6 Months	311
Sunglass	25	Plastic+Square	210
Sunglass	24	Metal+Oval	207
Frame	9	Plastic+Round	202
Sunglass	29	Plastic+Oval	202
Frame	5	Metal+Pilot	202
Frame	8	Plastic+Rectangle	198
Sunglass	19	Metal+Square	195
Frame	6	Metal+Oval	191
Sunglass	21	Metal+Round	190
Sunglass	20	Metal+Rectangle	189
Sunglass	27	Plastic+Round	177
Sunglass	26	Plastic+Rectangle	176
Frame	4	Metal+Hexagon	176
Frame	3	Metal+Round	174
Frame	10	Plastic+Hexagon	174
Frame	1	Metal+Square	173
Frame	11	Plastic+Oval	172



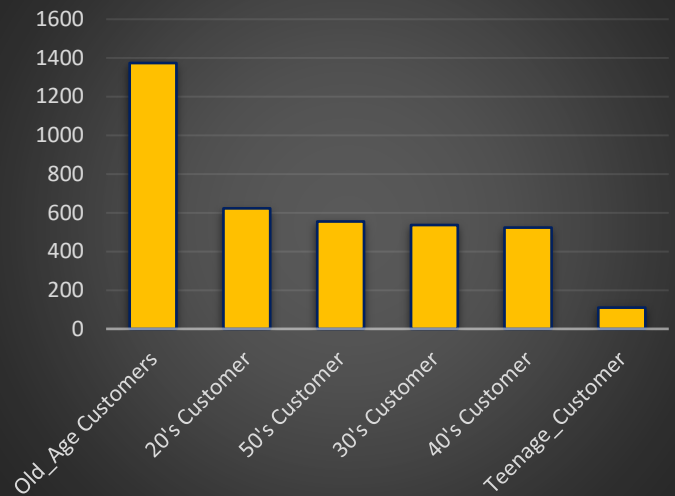
Dashboard Actual Buyer Customers

Share of Buyer Customers out of all the customers

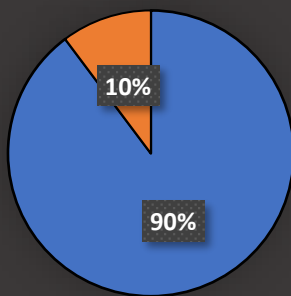


■ Total Visted Customers ■ Unique Customers

Buyer Customers

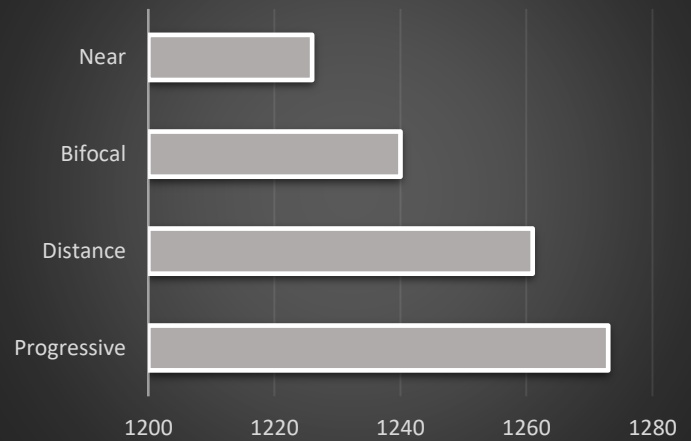


Buyer Customers across different customers type

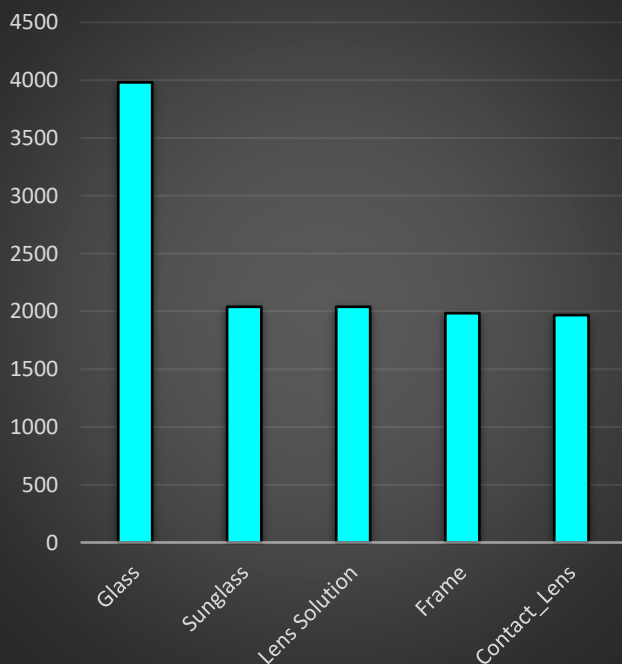


■ New
■ Referral

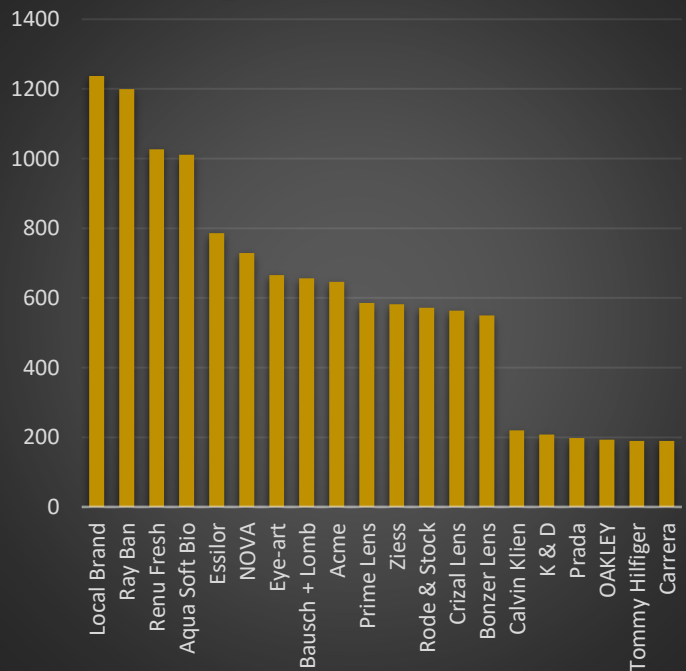
Buyers Customers v/s Vision Type



Quantity of Products Sold



QTY_SOLD by each Brand



• Repeated Customers

```
CREATE VIEW repeated_customers AS
SELECT i.customer_id
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
WHERE i.customer_id IS NOT NULL
GROUP BY i.customer_id
HAVING COUNT(*) > 1
ORDER BY COUNT(*) DESC;
```

The Repeated Customers VIEW was created to avoid rewriting the repeated customer identification logic across multiple queries. Once defined, this virtual table is reused across the Age Group, Vision Type, Revenue, Staff, and Product analysis modules making the queries cleaner, shorter, and easier to maintain. This is a standard best practice in production SQL environments.

Business Questions Answer about the Actual Buyer Customers are as follows:

- 1) How many customers have returned to make more than one purchase, and what percentage of our unique buyer base do repeated customers represent?
- 2) Which age group has the highest number of repeated customers, and does the age group that visits the most also show the strongest loyalty to the store?
- 3) How much total revenue has been generated exclusively by repeated customers, and what percentage of overall store revenue can be attributed to customer loyalty?
- 4) Which vision type is most commonly associated with repeated customers and does the nature of a customer's eye condition make them more likely to return?
- 5) What does the complete customer summary reveal total walk-ins, total buyers, unique buyers, conversion rate, repeated customers, and customer retention rate and what story do these numbers tell about the overall health of the business?
- 6) Which staff members have the highest number of repeated customers assigned to them who is building the strongest customer relationships?
- 7) Which repeat customers have not visited the store for the longest time, and how many days have passed since their last purchase?
- 8) How long have repeat customers remained active with the business, measured by the number of days between their first and most recent purchase?

Why this analysis matters

Repeated customers are one of the strongest indicators of customer satisfaction, trust, and long-term business sustainability. Acquiring a new customer is generally more expensive than retaining an existing one, making customer retention a critical driver of profitability. Analysing repeat customers helps identify the segments most likely to return, evaluate customer loyalty, and uncover opportunities to strengthen retention strategies, improve lifetime value, and build lasting customer relationships

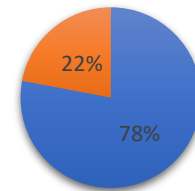
1) Repeated Customer Retention Rate

1) How many customers have returned to make more than one purchase, and what percentage of our unique buyer base do repeated customers represent?

SELECT

```
COUNT(DISTINCT i.customer_id) AS Total_Buyers,  
COUNT(DISTINCT rc.customer_id) AS Repeated_Customers,  
ROUND( COUNT(DISTINCT rc.customer_id) * 100 /  
COUNT(DISTINCT i.customer_id), 2)  
AS Retention_Rate  
FROM invoices as i  
LEFT JOIN repeated_customers as rc  
ON rc.customer_id=i.customer_id;
```

Share of Repeated Customers



■ Total Buyer Customer ■ Repeated Customers

Result Grid Filter Rows: Exports

	Total_Buyers	Repeated_Customers	Retention_Rate
▶	3721	1051	28.25

2) Count of Repeated Customers across different Age Groups

2) Which age group has the highest number of repeated customers, and does the age group that visits the most also show the strongest loyalty to the store?

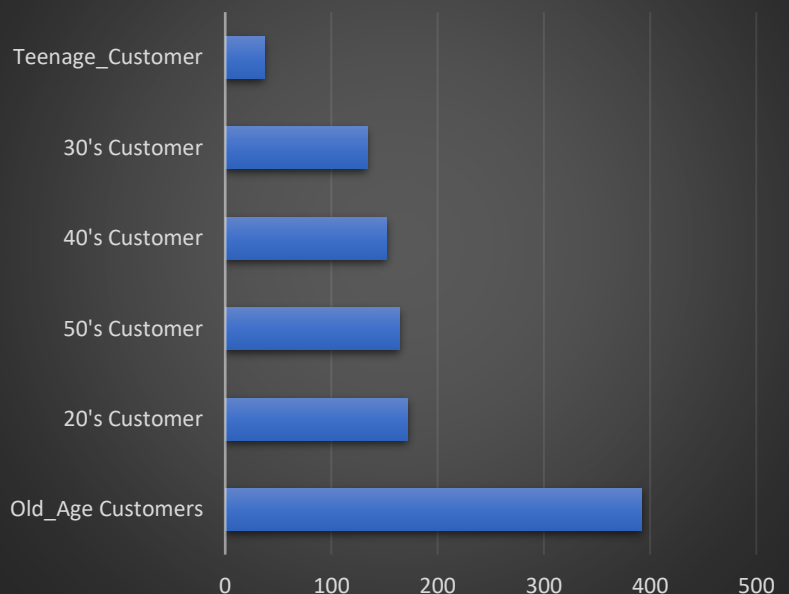
SELECT

```
CASE  
WHEN c.age < 20 THEN "Teenage_Customer"  
WHEN c.age BETWEEN 20 AND 30 THEN "20's Customer"  
WHEN c.age BETWEEN 31 AND 40 THEN "30's Customer"  
WHEN c.age BETWEEN 41 AND 50 THEN "40's Customer"  
WHEN c.age BETWEEN 51 AND 60 THEN "50's Customer"  
WHEN c.age > 60 THEN "Old_Age Customers"  
END AS Age_group,  
COUNT(*) AS Count_Repeated_Customers  
FROM customers_details as c  
INNER JOIN repeated_customers as rc  
ON c.customer_id=rc.customer_id  
GROUP BY Age_group  
ORDER BY Count_Repeated_Customers DESC;
```

Result Grid Filter Rows: E

	Age_group	Count_Repeated_Customers
▶	Old_Age Customers	392
	20's Customer	172
	50's Customer	164
	40's Customer	152
	30's Customer	134
	Teenage_Customer	37

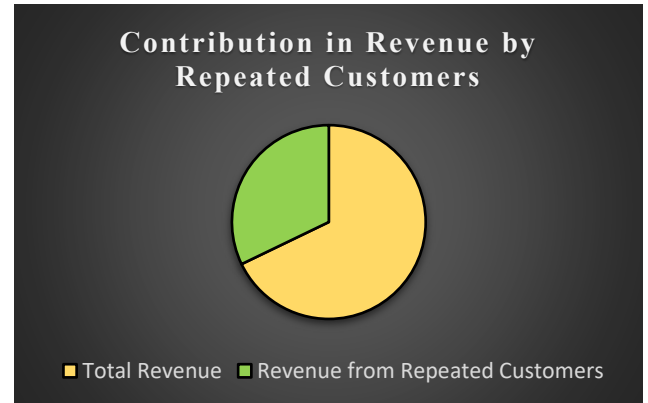
Repeated Customers across different age groups



3) Revenue Collected from Repeated Customers

3) How much total revenue has been generated exclusively by repeated customers, and what percentage of overall store revenue can be attributed to customer loyalty?

```
SELECT
SUM(i.final_bill_amount) AS Total_Revenue_Collected,
SUM(CASE WHEN rc.customer_id IS NOT NULL THEN i.final_bill_amount END)
AS Revenue_from_Repeated_Customers,
ROUND( SUM(CASE WHEN rc.customer_id IS NOT NULL THEN i.final_bill_amount END) * 100 /
SUM(i.final_bill_amount), 2)
AS Contribution_in_Revenue_by_Repeated_Customers
FROM invoices as i
LEFT JOIN repeated_customers as rc
ON i.customer_id=rc.customer_id;
```



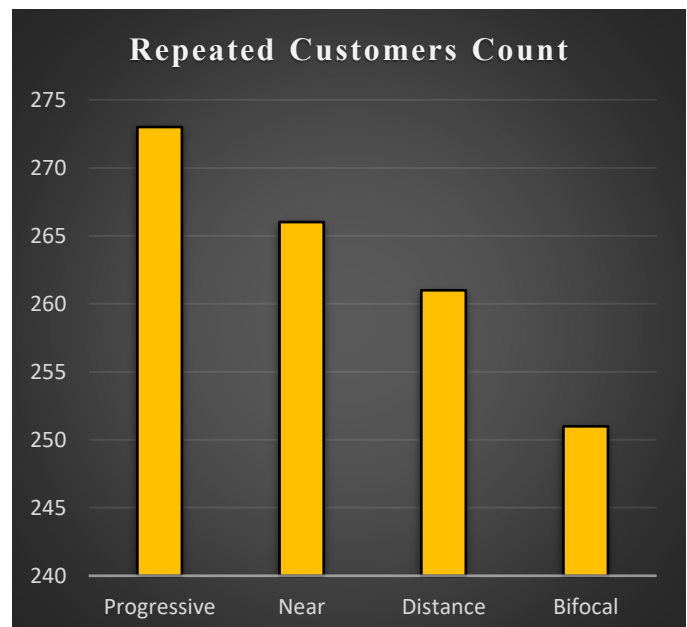
Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Total_Revenue_Collected	Revenue_from_Repeated_Customers	Contribution_in_Revenue_by_Repeated_Customers	
46811413.00	22178896.00	47.38	

4) Vision Type of Repeated Customers

4) Which vision type is most commonly associated with repeated customers and does the nature of a customer's eye condition make them more likely to return?

```
SELECT
p.vision_type, COUNT(rc.customer_id)
AS COUNT_Repeared_Customers
FROM prescriptions as p
INNER JOIN repeated_customers as rc
ON p.customer_id=rc.customer_id
GROUP BY p.vision_type
ORDER BY COUNT_Repeared_Customers DESC;
```

Result Grid	Filter Rows:
vision_type	COUNT_Repeared_Customers
Progressive	273
Near	266
Distance	261
Bifocal	251



5) Summary of Customers (Visited, Buyers, Repeated)

5) What does the complete customer summary reveal total walk-ins, total buyers, unique buyers, conversion rate, repeated customers, and customer retention rate and what story do these numbers tell about the overall health of the business?

```
SELECT
    COUNT( DISTINCT c.customer_id) AS Total_Visited_Customers,
    COUNT( i.customer_id) AS Total_Buyers,
    COUNT( DISTINCT i.customer_id) AS Total_Unique_Buyers,

    ROUND( COUNT(DISTINCT i.customer_id) * 100 /
    COUNT(DISTINCT c.customer_id), 2) AS Conversion_Rate,

    COUNT( DISTINCT rc.customer_id) AS Unique_Repeated_Customers,

    ROUND( COUNT( DISTINCT rc.customer_id) * 100 /
    COUNT( DISTINCT i.customer_id), 2) AS Customer_Retention_Rate
FROM customers_details as c
LEFT JOIN invoices as i
ON c.customer_id=i.customer_id
LEFT JOIN repeated_customers as rc
ON c.customer_id=rc.customer_id;
```

	Total_Visited_Customers	Total_Buyers	Total_Unique_Buyers	Conversion_Rate	Unique_Repeated_Customers	Customer_Retention_Rate
	8000	5000	3721	46.51	1051	28.25

6) Staff with highest number of customers

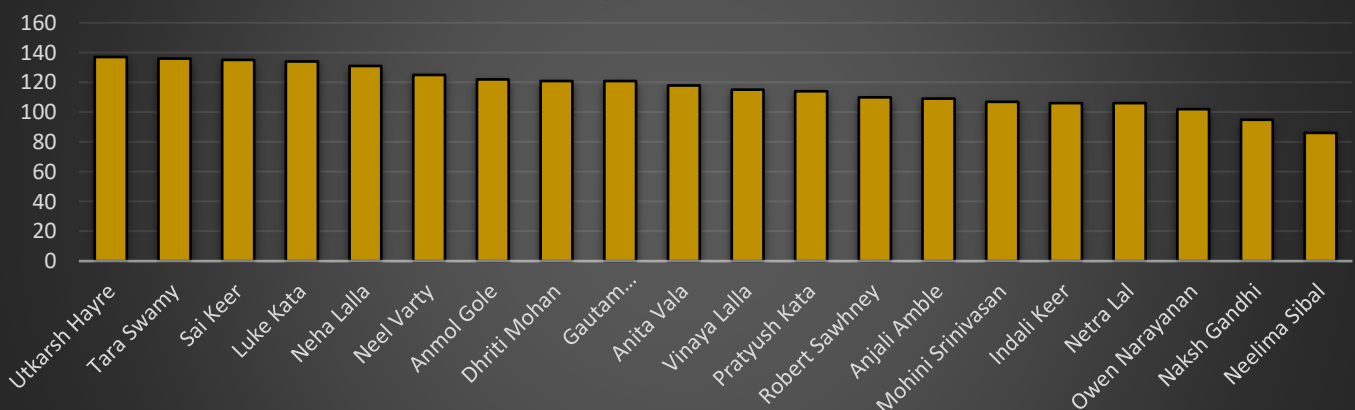
6) Which staff members have the highest number of repeated customers assigned to them who is building the strongest customer relationships?

```
WITH repeated_customers_staff_id AS (
SELECT rc.customer_id, c.assigned_staff_id
FROM repeated_customers as rc
INNER JOIN customers_details as c
ON rc.customer_id=c.customer_id
),
repeated_customers_staff_name AS (
SELECT
    rcsi.customer_id, rcsi.assigned_staff_id,
    s.staff_full_name
FROM repeated_customers_staff_id as rcsi
INNER JOIN staff as s
ON s.staff_id=rcsi.assigned_staff_id
),
```

```
repeated_customers_staff_invoices as (
SELECT
    rcsn.staff_full_name,
    rcsn.assigned_staff_id,
    rcsn.customer_id, i.invoice_id
FROM repeated_customers_staff_name as rcsn
INNER JOIN invoices as i
ON rcsn.customer_id=i.customer_id
)
SELECT
    rcsni.staff_full_name,
    COUNT(rcsni.customer_id)
FROM repeated_customers_staff_invoices as rcsni
GROUP BY rcsni.staff_full_name WITH ROLLUP
ORDER BY COUNT(rcsni.customer_id) DESC;
```

	staff_full_name	Repeated_Customers
	Utkarsh Hayre	2330
	Tara Swamy	137
	Sai Keer	136
	Luke Kata	135
	Neha Lalla	134
	Neel Varty	131
	Anmol Gole	125
	Dhriti Mohan	122
	Gautam Choudhary	121
	Anita Vala	118
	Vinaya Lalla	115
	Pratyush Kata	114
	Robert Sawhney	110
	Anjali Amble	109
	Mohini Srinivasan	107
	Indali Keer	106
	Netra Lal	106
	Owen Narayanan	102
	Naksh Gandhi	95
	Neelima Sibal	86

Staff with count of Repeated Customers



7) Identify repeated customers who are at risk of churning

7) Which repeat customers have not visited the store for the longest time, and how many days have passed since their last purchase?

```
WITH churning_customers AS (
SELECT
    rc.customer_id,
    MAX(i.invoice_date) AS Last_Visit_Date,
    DATEDIFF(CURDATE(), MAX(i.invoice_date))
    AS Days_from_last_visit
FROM repeated_customers as rc
INNER JOIN invoices as i
ON rc.customer_id=i.customer_id
GROUP BY rc.customer_id
HAVING Days_from_last_visit > 180
ORDER BY Days_from_last_visit DESC
),
churning_customers_details AS (
SELECT
    cc.customer_id, c.customer_full_name,
    c.mobile, c.age, cc.Last_Visit_Date,
    cc.Days_from_last_visit
FROM churning_customers as cc
INNER JOIN customers_details as c
ON cc.customer_id=c.customer_id
)
SELECT
    ccd.customer_id, ccd.customer_full_name,
    ccd.mobile, ccd.age, ccd.Last_Visit_Date,
    ccd.Days_from_last_visit
FROM churning_customers_details as ccd;
```

customer_id	customer_full_name	mobile	age	Last_Visit_Date	Days_from_last_visit
7325	Saumya Karpe	0568512814	49	2025-01-26 00:00:00	517
3293	Vedant Saxena	5480227836	22	2025-02-06 00:00:00	506
4169	Hitesh Korpai	1624156656	42	2025-02-06 00:00:00	506
1166	Quincy Bala	7314923403	51	2025-02-07 00:00:00	505
3944	Eesha Banik	2033560086	38	2025-02-10 00:00:00	502
3986	Maanas Brar	1643364188	22	2025-02-11 00:00:00	501
6266	Mugdha Datta	1866781344	34	2025-02-13 00:00:00	499
1809	Leena Dugal	9621641944	76	2025-02-16 00:00:00	496
1111	Krish Bail	0183544535	60	2025-02-19 00:00:00	493
6616	Kala Vora	2127629587	45	2025-02-19 00:00:00	493
1629	Niharika Nath	6059400213	75	2025-02-20 00:00:00	492
6708	Caleb Konda	9203467054	65	2025-02-22 00:00:00	490
7005	Chasmum Wali	0166012578	53	2025-02-22 00:00:00	490
2325	Anvi Tara	3461214400	30	2025-02-24 00:00:00	488
7329	Indrajit Devan	1076196510	75	2025-02-25 00:00:00	487
4664	Jagrati Narula	7546971956	51	2025-02-26 00:00:00	486
344	Kiaan Madan	5929300620	23	2025-02-27 00:00:00	485
2764	Chakradev Das	3541300349	62	2025-02-27 00:00:00	485
4500	Dominic Kothari	1735730331	53	2025-02-27 00:00:00	485
1867	Madhav Palan	2414597827	30	2025-03-01 00:00:00	483
7522	Amruta Minhas	7940878896	33	2025-03-07 00:00:00	477
1000	Sai Borra	6087717856	18	2025-03-08 00:00:00	476
1558	Jackson Pall	5004065825	79	2025-03-08 00:00:00	476
1263	Ayush Bhatnagar	1360567281	60	2025-03-11 00:00:00	473
1105	Varsha Lalla	2052977582	35	2025-03-12 00:00:00	472
2194	Oscar Krish	1602062656	50	2025-03-13 00:00:00	471
3133	Baljiwan Sant	6779228506	24	2025-03-13 00:00:00	471
2012	Robert Babu	2654246361	80	2025-03-14 00:00:00	470
832	Aarini Dugar	2445055763	52	2025-03-18 00:00:00	466
7159	Ekantika Ravel	0884976467	75	2025-03-18 00:00:00	466

8) Evaluate the repeated customers loyalty and longevity, how long customer is associated with the business

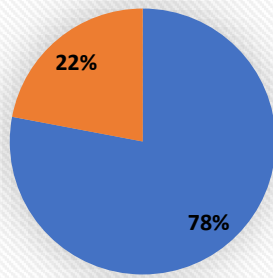
8) How long have repeat customers remained active with the business, measured by the number of days between their first and most recent purchase?

```
WITH loyal_customers AS (
SELECT
    rc.customer_id,
    DATE(MIN(i.invoice_date)) AS First_Purchase_Date,
    DATE(MAX(i.invoice_date)) AS Last_Purchase_Date,
    DATEDIFF(MAX(i.invoice_date), MIN(i.invoice_date))
    AS Days_Between_First_and_Last_Purchase
FROM repeated_customers AS rc
INNER JOIN invoices AS i
ON rc.customer_id = i.customer_id
GROUP BY rc.customer_id
ORDER BY Days_Between_First_and_Last_Purchase DESC
),
loyal_customers_details AS (
SELECT
    lc.customer_id,
    c.customer_full_name,
    c.age,
    c.mobile,
    lc.First_Purchase_Date,
    lc.Last_Purchase_Date,
    lc.Days_Between_First_and_Last_Purchase
FROM loyal_customers AS lc
INNER JOIN customers_details AS c
ON lc.customer_id=c.customer_id
)
SELECT lcd.customer_id,
    lcd.customer_full_name,
    lcd.age,
    lcd.mobile,
    lcd.First_Purchase_Date,
    lcd.Last_Purchase_Date,
    lcd.Days_Between_First_and_Last_Purchase
FROM loyal_customers_details as lcd;
```

customer_id	customer_full_name	age	mobile	First_Purchase_Date	Last_Purchase_Date	Days_Between_First_and_Last_Purchase
4446	Fitan Kala	77	0905725157	2025-01-11	2026-01-08	362
6749	Rachita Trivedi	49	4796779747	2025-01-16	2026-01-05	354
5318	Daniel Goel	81	5662460088	2025-01-19	2026-01-07	353
5321	Devika Sule	26	5439696076	2025-01-15	2026-01-02	352
5011	Chakradev Kalla	72	3972427488	2025-01-19	2026-01-03	349
2727	Fitan Murty	50	3555635597	2025-01-20	2025-12-28	342
1037	Siya Tiwari	45	8468277430	2025-01-17	2025-12-21	338
557	Vinda Aggarwal	28	5390263719	2025-01-23	2025-12-24	335
2810	Yoshita Narang	53	9801756787	2025-01-22	2025-12-23	335
3402	Ati Deol	69	3883301859	2025-01-12	2025-12-09	331
4563	Varsha Bains	84	1212355200	2025-02-05	2026-01-01	330
2303	Hitesh Rout	74	7922069089	2025-02-06	2026-01-02	330
1790	Vrishti Chauhan	18	8391274033	2025-02-07	2026-01-02	329
3682	Adweta Bhatti	85	2755483105	2025-02-05	2025-12-29	327
4330	Samaksh Ganesh	61	4260011729	2025-01-18	2025-12-10	326
4542	Wriddhish Tailor	73	2643285721	2025-01-22	2025-12-12	324
5765	Yashawini Bhat	38	9687857802	2025-01-16	2025-12-06	324
6591	Yashica Prabhu	54	7424093041	2025-02-21	2026-01-09	322
7976	Odika Murthy	59	9047291834	2025-01-29	2025-12-17	322
3458	Lila Zachariah	24	3018456384	2025-01-25	2025-12-13	322
5561	Mekhala Boase	71	3805756164	2025-02-01	2025-12-19	321
6914	Champak Oza	54	5210065648	2025-01-12	2025-11-29	321
3922	Amara Badal	21	5403354613	2025-01-19	2025-12-05	320
5025	Daniel Karnik	59	0928907098	2025-01-23	2025-12-08	319
6182	Devika Mallick	32	1900216102	2025-01-12	2025-11-27	319
1800	Bimala Hari	55	7786840732	2025-01-12	2025-11-26	318
7858	Shivansh Maharaj	19	9057874059	2025-01-27	2025-12-11	318
4011	Ikshta Dhaliwal	62	1699751299	2025-03-01	2026-01-11	316
382	Prisha Dhillon	35	0089727123	2025-01-18	2025-11-29	315
4959	Vasudha Taneja	19	5891565757	2025-02-16	2025-12-26	313
4216	Kritika Goyal	29	7086022525	2025-01-24	2025-12-03	313

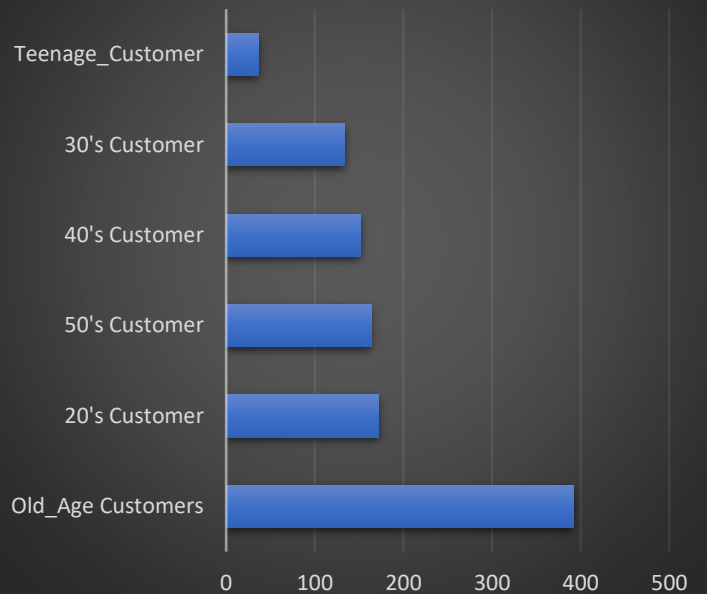
Repeated Customers Dashboard

Share of Repeated Customers

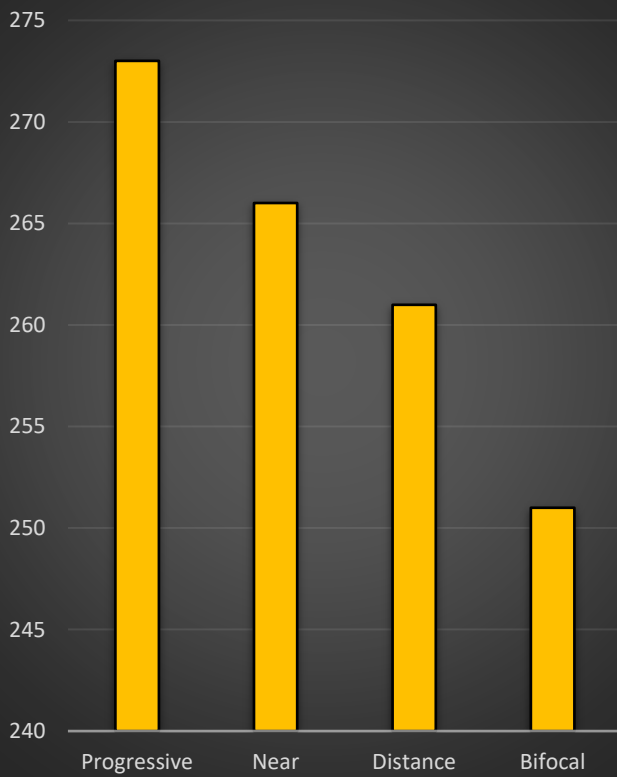


■ Total Buyer Customer ■ Repeated Customers

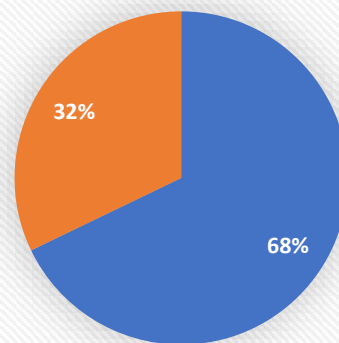
Repeated Customers across different age groups



Repeated Customers Count



Contribution in Revenue by Repeated Customers



■ Total Revenue ■ Revenue from Repeated Customers

Staff with count of Repeated Customers

